

GOLDSBORO-WAYNE, NC, USA

WMO: 746940

Lat: 35.460N

Lon: 77.965W

Elev: 41

StdP: 100.84

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 03708

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-3.8	-15.1	1.0	-1.5	-12.4	1.3	0.1	9.3	14.4	8.4	12.2	1.7	0	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	35.1	24.4	33.6	24.3	32.5	24.0	26.6	31.8	25.9	30.7	25.4	30.0	3.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.5	28.5	24.8	20.0	27.9	24.1	19.1	27.1	83.1	31.4	80.1	30.5	77.7	29.9	32.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
8.2	7.2	5.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-9.5	37.5	1.7	2.4	-10.7	39.3	-11.7	40.7	-12.6	42.0	-13.9	43.8
			-10.8	28.0	2.4	1.6	-12.5	29.1	-13.9	30.0	-15.2	30.9	-16.9	32.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	5.7	7.3	11.0	16.4	21.1	25.3	26.8	25.6	22.5	16.9	11.0	7.9	(6)
(7)		DBStd	8.55	5.87	5.23	5.30	4.57	3.92	2.96	2.06	2.27	2.87	4.40	4.76	5.09	(7)
(8)		HDD10.0	474	160	106	53	4	0	0	0	0	0	3	43	104	(8)
(9)		HDD18.3	1676	392	309	232	89	20	1	0	4	80	224	325	(9)	
(10)		CDD10.0	2856	27	32	86	198	344	459	520	484	377	217	74	39	(10)
(11)		CDD18.3	1013	1	2	6	32	106	209	262	226	130	35	4	2	(11)
(12)		CDH23.3	9081	1	9	56	301	943	2046	2556	1931	943	263	29	5	(12)
(13)		CDH26.7	3656	0	0	4	65	338	914	1163	803	308	57	2	0	(13)
(14)	Wind	WSAvg	2.3	2.5	2.7	2.8	2.9	2.5	2.2	2.0	1.5	1.9	2.1	2.1	(14)	
(15)	Precipitation	PrecAvg	1272	88	83	94	92	99	122	149	137	156	87	83	78	(15)
(16)		PrecMax	1781	186	189	156	166	219	327	313	374	726	179	199	168	(16)
(17)		PrecMin	908	22	36	22	29	45	39	69	44	27	0	8	15	(17)
(18)		PrecStd	211	46	42	38	40	44	65	61	66	134	47	50	40	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	25.0	27.4	30.7	34.7	36.4	37.0	35.9	33.1	31.0	27.0	23.8	(19)
(20)		MCWB	17.9	18.3	17.5	20.0	23.1	23.5	25.4	24.9	23.3	22.7	19.7	19.6	(20)	
(21)		2%	DB	20.7	21.9	25.0	28.4	32.2	34.9	35.0	33.7	32.0	28.0	23.7	21.3	(21)
(22)		MCWB	16.6	16.3	16.9	18.8	21.7	23.8	25.1	24.8	23.1	20.8	17.4	18.3	(22)	
(23)		5%	DB	17.6	19.2	22.6	27.1	30.3	33.0	33.4	32.4	30.1	26.2	21.6	19.3	(23)
(24)		MCWB	14.1	14.8	15.7	18.2	21.4	23.4	25.1	24.7	22.8	19.7	17.5	17.2	(24)	
(25)		10%	DB	15.2	16.8	20.2	25.0	28.2	32.0	32.4	31.2	28.1	24.3	19.8	17.1	(25)
(26)		MCWB	12.1	12.5	14.7	17.6	20.8	23.3	24.6	24.4	22.0	19.9	16.5	14.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.0	19.6	19.7	22.7	24.6	26.3	27.8	27.2	25.8	24.8	21.7	21.3	(27)
(28)		MADB	21.0	22.7	23.7	26.0	29.8	31.8	33.3	32.3	29.7	27.2	24.4	22.5	(28)	
(29)		2%	WB	17.2	18.0	18.5	21.2	23.6	25.5	26.8	26.5	25.1	23.4	20.1	19.1	(29)
(30)		MADB	19.3	20.6	22.6	24.7	28.6	31.1	32.2	31.4	29.0	25.7	21.7	20.5	(30)	
(31)		5%	WB	15.2	15.8	17.4	20.1	22.8	25.0	26.1	25.8	24.2	22.1	18.5	17.3	(31)
(32)		MADB	17.0	18.6	20.4	24.1	27.4	30.3	31.2	30.2	27.4	24.9	20.9	18.9	(32)	
(33)		10%	WB	12.8	13.1	16.2	18.9	22.0	24.3	25.5	25.2	23.4	20.6	16.7	15.0	(33)
(34)		MADB	14.8	15.8	19.6	23.4	26.5	29.3	30.3	29.2	26.3	22.9	19.2	16.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.3	11.2	12.3	12.9	11.8	11.3	10.9	10.3	10.5	11.8	12.3	10.7	(35)	
(36)		5% DB	MCDBR	13.2	14.2	14.5	14.5	13.4	12.8	12.4	11.8	12.5	13.2	13.7	12.2	(36)
(37)		MCWBR	9.8	9.5	7.8	6.4	4.8	4.1	4.4	4.5	5.2	6.5	8.5	9.1	(37)	
(38)		5% WB	MCDBR	11.7	13.4	12.1	11.6	11.1	11.0	11.1	10.1	9.7	9.8	10.9	11.2	(38)
(39)		MCWBR	10.1	10.4	8.1	6.4	4.6	4.1	4.6	4.4	4.7	5.7	8.1	9.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.332	0.360	0.406	0.457	0.503	0.537	0.525	0.421	0.370	0.333	0.316	(40)	
(41)		taud	2.492	2.440	2.393	2.268	2.161	2.078	1.999	2.023	2.313	2.433	2.494	2.525	(41)	
(42)		Ebn at Noon	888	910	909	879	835	793	763	764	835	854	859	864	(42)	
(43)		Edh at Noon	83	98	112	134	151	164	176	169	120	98	83	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.60	3.29	4.33	5.47	6.03	6.39	6.10	5.50	4.58	3.73	2.86	2.28	(44)	
(45)		RadStd	0.17	0.36	0.29	0.36	0.45	0.40	0.35	0.30	0.35	0.47	0.22	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+113

Nomenclature: See separate page